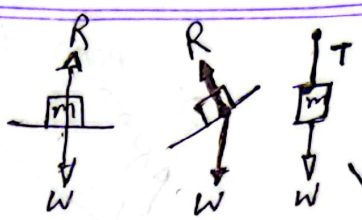
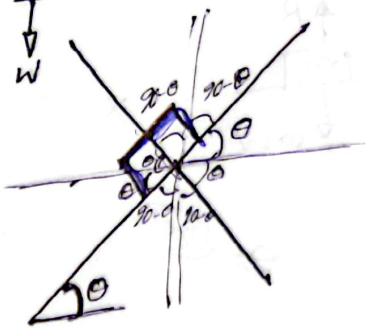
~~34~~



Weight → $W = mg$



Normal reaction force → R or F_N

$\boxed{\text{not always } R = mg}$

Tension → T

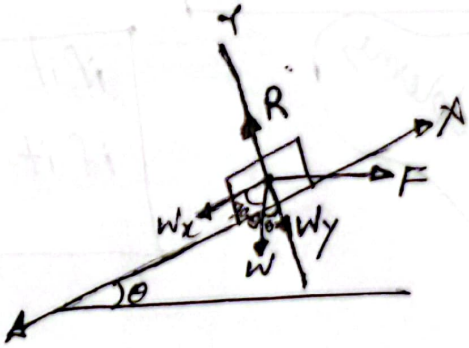
Frictional force → Static Friction → $f_s = \mu_s R$ $\boxed{\text{opposite to } F}$

Frictional force → Kinetic Friction → $f_k = \mu_k R$

34| $m = 100kg$

$\theta = 30°$

$W = mg = 100 \times 9.8 = 980$



$W \to (980, 240°)$

$R \to (R, 90°)$

$F \to (F_a, 380°)$

$a \to (0, 0)$

$\vec{W} = 980\cos 240°\,\hat{i} + 980\sin 240°\,\hat{j}$

$= -490\hat{i} - 848.7\hat{j}$

$\vec{R} = R\cos 90°\,\hat{i} + R\sin 90°\,\hat{j}$

$= 0\hat{i} + R\hat{j}$

$\vec{F} = F\cos 330^\circ \hat{i} + F\sin 330^\circ \hat{j}$

$= 0.86F\hat{i} - 0.5F\hat{j}$

$\vec{a} = 0\hat{i} + 0\hat{j}$

$\vec{F}_{net} = m\vec{a}$

$x \rightarrow F_{net,x} = ma_x$

$\Rightarrow -490 + 0 + 0.86F = 100 \times 0$

$\Rightarrow F = \dfrac{490}{0.86} = 569.7\,N$

$y \rightarrow F_{net,y} = ma_y$

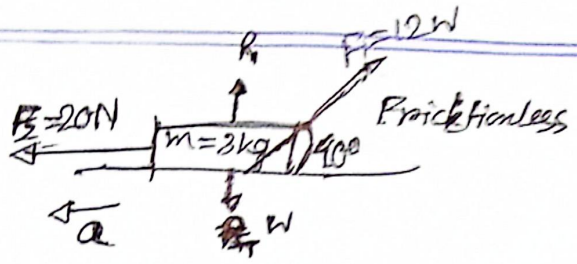
$\Rightarrow -848.7 + R - 0.5F = 100 \times 0$

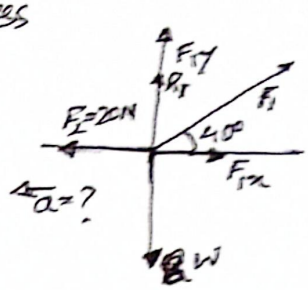
$\Rightarrow R = 848.7 + (0.5 \times 569.7)$

$\Rightarrow R = 1133.6\,N$

28 Feb

Quiz-2 Phy





$\vec{F_1} = 12\cos 40°\,\hat{i} + 12\sin 40°\,\hat{j}$

$= 9.19\,\hat{i} + 7.71\,\hat{j}$

$\vec{R_1} = 29.4\,\hat{j}$ $\quad (\because W = -29.4\,\hat{j})$

$\vec{F_2} = -20\,\hat{i}$

$\vec{F'} = 9.19\,\hat{i} - 20\,\hat{i}$

$= -10.81\,\hat{i}$

~~$|F'| = ma$~~

~~$\Rightarrow 10.81 = 3a$~~

~~$\Rightarrow a = 3.6\ ms^{-2}$~~

~~$R' = 29.4\,\hat{j} + 7.71\,\hat{j}$~~

~~$= 37.11\ N$~~

$\vec{F'} = m\vec{a}$

$\Rightarrow -10.81\,\hat{i} = 3\vec{a}$

$\Rightarrow \vec{a} = -3.6\,\hat{i}\ ms^{-2}$

$y \rightarrow F_{net,y} = ma_y$

$\Rightarrow -29.4 + R_1 + 7.71 + 0 = 3 \times 0$

$\therefore R_1 = 21.7N$